

13-subgroup: 子群 (Subgroup)

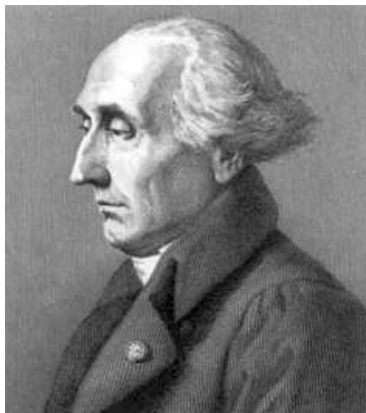
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Lagrange's Theorem



Joseph-Louis Lagrange (1736 ~ 1813)

Fundamental Homomorphism Theorem



Emmy Noether (1882 ~ 1935)

Lagrange's Theorem

Help us understand the structure of a group
via its subgroups/normal subgroups

Fundamental Homomorphism Theorem

Definition (Symmetric Group (对称群; $\text{Sym}(M)$))

Let $M \neq \emptyset$ be a set.

All the **permutations/bijections** of M , together with the **composition** operation, is a group, called the **symmetric group** of M .

$$M = \{1, 2, \dots, n\}$$

$$S_n \triangleq \text{Sym}(M)$$

$$S_3$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \quad \tau = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

$$\sigma\tau = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$

$$\tau\sigma = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

$$\sigma\tau \neq \tau\sigma$$

Cyclic Notation (轮换表示法) & Transposition (对换)

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 4 & 3 & 6 & 1 & 5 & 2 \end{pmatrix}$$

$$\sigma = (1\ 4)(2\ 3\ 6)(5)$$

$$= (1\ 4)(2\ 3\ 6)$$

$$= (2\ 3\ 6)(1\ 4)$$

$$= (2\ 3\ 6)(4\ 1)$$

$$= (3\ 6\ 2)(4\ 1)$$

$$= (3\ 6)(6\ 2)(4\ 1)$$

$$(i_1\ i_2\ \dots\ i_r) = (i_1\ i_2)(i_2\ i_3)\dots(i_{r-2}\ i_{r-1})(i_{r-1}\ i_r)$$

By induction on the length r .

$$S_3$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

$$(1) \quad (1\ 2) \quad (1\ 3) \quad (2\ 3) \quad (1\ 3\ 2) \quad (1\ 2\ 3)$$

Definition (Permutation Group (置换群))

Let $M \neq \emptyset$ be a set.

A **permutation group** of M is a **subgroup** of $\text{Sym}(M)$.

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2)\} \leq S_3$$

$$H = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \leq S_3$$

$$\begin{aligned}\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 7 & 3 & 6 & 2 & 5 & 4 & 1 \end{pmatrix} &= (1\ 7)(2\ 3)(3\ 6)(6\ 4) \\ &= (1\ 7)(3\ 6)(2\ 5)(6\ 4)(4\ 5)(2\ 5)\end{aligned}$$

Theorem (Parity (奇偶性) of Permutations)

将一个置换表示成若干对换的乘积, 所用对换个数的奇偶性是唯一的。

Definition (Even/Odd Permutations (偶置换/奇置换))

可表示为偶数个对换的乘积的置换称为偶置换; 否则, 称为奇置换。

Definition (Alternating Group (交错群; A_n))

由 S_n 的全体偶置换构成的子群称为 n 次交错群。

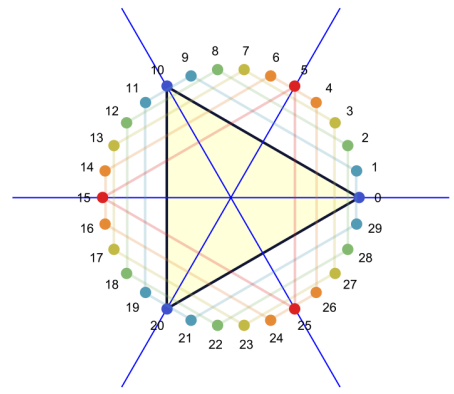
$$A_3 = \{(1), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$\operatorname{sgn} : S_n \rightarrow \{1, -1\}$$

$$\operatorname{sgn}(\sigma) = \begin{cases} 1 & \sigma \in A_n, \\ -1 & \sigma \notin A_n \end{cases}$$

$$\operatorname{sgn}(\sigma_1\sigma_2) = \operatorname{sgn}(\sigma_1)\operatorname{sgn}(\sigma_2)$$

$$S_n/A_n \cong \{1, -1\}$$



$$A_3 = \{(1), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$(1\ 2)A_3 = \{(1\ 2), (2\ 3), (1\ 3)\}$$

Definition (Coset (陪集))

Suppose that $H \leq G$. For $a \in G$,

$$aH = \{ah \mid h \in H\}, \quad Ha = \{ha \mid h \in H\},$$

is called the **left coset** (左陪集) and **right coset** of H in G , respectively.

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2)\} \leq S_3$$

$$(1)H = H = (1\ 2)H \leq S_3$$

$$(1\ 3)H = \{(1\ 3), (1\ 2\ 3)\} = (1\ 2\ 3)H$$

$$(2\ 3)H = \{(2\ 3), (1\ 3\ 2)\} = (1\ 3\ 2)H$$

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \leq S_3$$

$$(1)H = (1\ 2\ 3)H = (1\ 3\ 2)H = H \leq S_3$$

$$(1\ 2)H = (1\ 3)H = (2\ 3)H = \{(1\ 2), (1\ 3), (2\ 3)\}$$

Theorem

Suppose that $H \leq G$, $a, b \in G$.

(1)

$$|aH| = |H| = |bH|$$

(2)

$$a \in aH$$

(3)

$$aH = H \iff a \in H \iff aH \leq G$$

(4)

$$aH = bH \iff a^{-1}b \in H$$

(5)

$$\forall a, b \in G. (aH = bH) \vee (aH \cap bH = \emptyset)$$

$$aH = bH \iff a^{-1}b \in H$$

$$\boxed{a^{-1}b \in H \iff a^{-1}bH = H}$$

$$aH = bH \implies a^{-1}aH = a^{-1}bH \implies a^{-1}bH = H \implies a^{-1}b \in H$$

$$a^{-1}bH = H \implies a(a^{-1}bH) = aH \implies bH = aH$$

$$\forall a, b \in G. (aH = bH) \vee (aH \cap bH = \emptyset)$$

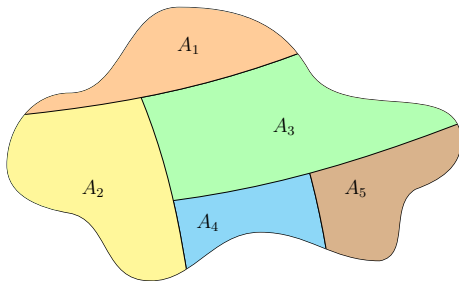
$$\forall a, b \in G. (aH \cap bH \neq \emptyset \rightarrow aH = bH)$$

Take any $g \in aH \cap bH$.

$$\exists h_1, h_2 \in H. (ah_1 = g = ah_2) \wedge (h_1H = H = h_2H)$$

$$aH = a(h_1H) = (ah_1)H = (bh_2)H = b(h_2H) = bH$$

A balanced partition of G by its subgraph H



Theorem (Lagrange's Theorem)

Suppose that $H \leq G$. Then

$$|G| = [G : H] \cdot |H|$$

Definition (Index (指标))

$$G/H = \{gH \mid g \in G\}$$

$$[G : H] \triangleq |G/H|$$

$$H \leq G \implies |H| \mid |G|$$

There are *no* subgroups of order 5, 7, or 8 of a group of order 12.

Theorem

- ▶ *There are only 2 groups of order 4.*
- ▶ *There are only 2 groups of order 6.*

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2)\} \leq S_3$$

$$(1)H = H = (1\ 2)H$$

$$(1\ 3)H = \{(1\ 3), (1\ 2\ 3)\} = (1\ 2\ 3)H$$

$$(2\ 3)H = \{(2\ 3), (1\ 3\ 2)\} = (1\ 3\ 2)H$$

$$H(1) = H = H(1\ 2)$$

$$H(1\ 3) = \{(1\ 3), (1\ 3\ 2)\} = (1\ 3\ 2)H$$

$$H(2\ 3) = \{(2\ 3), (1\ 2\ 3)\} = (1\ 2\ 3)H$$

It is possible that $aH \neq Ha$.

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \leq S_3$$

$$(1)H = (1\ 2\ 3)H = (1\ 3\ 2)H = H \leq S_3$$

$$(1\ 2)H = (1\ 3)H = (2\ 3)H = \{(1\ 2), (1\ 3), (2\ 3)\}$$

$$\forall a \in S_3. aH = Ha$$

Definition (Normal Subgroup (正规子群))

Suppose that $H \leq G$. If

$$\forall a \in G. aH = Ha,$$

then H is a **normal subgroup** of G , denoted $H \triangleleft G$.

$$aH = Ha \not\Rightarrow \forall h \in H. ah = ha$$

$$aH = Ha \Rightarrow \forall h \in H. \exists h' \in H. ah = h'a$$

Theorem

$$H \triangleleft G \iff \forall a \in G, h \in H. aha^{-1} \in H$$

$$\begin{aligned} aH = Ha &\implies aHa^{-1} = (Ha)a^{-1} = H(aa^{-1}) = H \\ &\implies aHa^{-1} \subseteq H \\ &\implies \forall h \in H. aha^{-1} \in H \end{aligned}$$

$$aha^{-1} \in H \implies ah = (aha^{-1})a \in Ha \implies aH \subseteq Ha$$

$$a^{-1}ha = a^{-1}h(a^{-1})^{-1} \in H \implies ha \in aH \implies Ha \subseteq aH$$

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \triangleleft S_3$$

$$\forall \sigma \in S_3, \tau \in H. \sigma\tau\sigma^{-1} \in H$$

Theorem

$$\sigma\tau\sigma^{-1} = \begin{pmatrix} \sigma(1) & \sigma(2) & \dots & \sigma(n) \\ \sigma(\tau(1)) & \sigma(\tau(2)) & \dots & \sigma(\tau(n)) \end{pmatrix}$$

$$(1\ 2)(1\ 2\ 3)(1\ 2)^{-1} = \begin{pmatrix} 2 & 1 & 3 \\ 1 & 3 & 2 \end{pmatrix} = (1\ 3\ 2)$$

Definition (正规子群的陪集)

Suppose that $H \triangleleft G$.

$$G/H = \{aH \mid a \in G\}$$

is the **set of cosets** of H in G .

Definition (Quotient Group (商群))

Suppose that $H \triangleleft G$. Define

$$aH \cdot bH = (ab)H.$$

Then $(G/H, \cdot)$ is a group, called the **quotient group** of G by H (denoted G/H).

$aH \cdot bH = (ab)H$ is well-defined

$$aH = a'H \wedge bH = b'H \implies aH \cdot bH = a'H \cdot b'H$$

结果与代表元的选取无关

$$S_3 = \{(1), (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$H = \{(1), (1\ 2\ 3), (1\ 3\ 2)\} \triangleleft S_3$$

$$G/H = \{(1)H, (1\ 2)H\}$$

$$G = \mathbb{Z} \quad H = 6\mathbb{Z} \triangleleft G$$

$$G/H = \{0 + H, 1 + H, \dots, 5 + H\}$$

Definition (Homomorphism (同态))

Let $(G, \cdot)$ and $(G', *)$ be two groups. Let ϕ be a function such that

$$\forall a, b \in G. \phi(ab) = \phi(a)\phi(b).$$

Then ϕ is a **homomorphism** from G to G' .

If ϕ is a bijection, then G and G' are called **isomorphic**.

$$\phi : G \cong G'$$

$$\phi : \mathbb{Z} \rightarrow \mathbb{R}^*$$

$$n \mapsto (-1)^n$$

$$\phi(m+n) = (-1)^{m+n} = \phi(m)\phi(n)$$

$$\phi : \mathbb{Z} \rightarrow \mathbb{Z}_6$$

$$a \mapsto [a]_6$$

$$\phi(a + b) = [a + b]_6 = \phi(a) + \phi(b)$$

$$\begin{aligned}\phi : \mathbb{R}[x] &\rightarrow \mathbb{R}[x] \\ f(x) &\mapsto f'(x)\end{aligned}$$

$\mathbb{R}[x]$: 全体实系数多项式关于多项式的加法构成的群

$$\phi(f(x) + g(x)) = (f(x) + g(x))' = \phi(f(x)) + \phi(g(x))$$

Theorem

Suppose that ϕ is a homomorphism from G to G' .

Let e and e' be identities of G and G' , respectively.

$$(1) \quad \phi(e) = e'$$

$$(2) \quad \phi(a^{-1}) = (\phi(a))^{-1}$$

$$e'\phi(e) = \phi(e) = \phi(ee) = \phi(e)\phi(e) \implies \phi(e) = e'$$

$$\phi(a)\phi(a^{-1}) = \phi(aa^{-1}) = \phi(e) = e' = \phi(a)(\phi(a))^{-1}$$

Theorem

Suppose that ϕ is a homomorphism from G to G' .

(1)

$$H \leq G \implies \phi(H) \leq G'$$

(2)

$$H \triangleleft G \implies \phi(H) \triangleleft \phi(G)$$

(3)

$$K \leq G' \implies \phi^{-1}(K) \leq G$$

(4)

$$K \triangleleft G' \implies \phi^{-1}(K) \triangleleft G$$

Definition (核 (Kernel))

Suppose that ϕ is a homomorphism from G to G' .

Let e' be the identity of G' .

$$\phi^{-1}(\{e'\}) = \{a \in G \mid \phi(a) = e'\}$$

is the **kernel** of ϕ , denoted $\text{Ker } \phi$.

$$\text{Ker } \phi \triangleleft G$$

$$\phi : \mathbb{Z} \rightarrow \mathbb{R}^*$$

$$n \mapsto (-1)^n$$

$$\text{Ker } \phi = 2\mathbb{Z}$$

$$\phi : \mathbb{Z} \rightarrow \mathbb{Z}_6$$

$$a \mapsto [a]_6$$

$$\text{Ker } \phi = 6\mathbb{Z}$$

$$\begin{aligned}\phi : \mathbb{R}[x] &\rightarrow \mathbb{R}[x] \\ f(x) &\mapsto f'(x)\end{aligned}$$

$\mathbb{R}[x]$: 全体实系数多项式关于多项式的加法构成的群

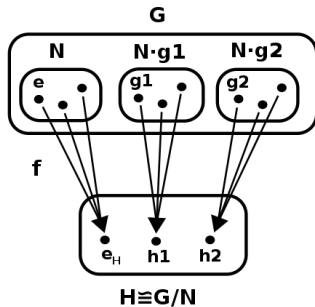
$$\text{Ker } \phi = \mathbb{R}$$

Theorem (Fundamental Homomorphism Theorem)

Suppose that ϕ is a homomorphism from G to G' . Then

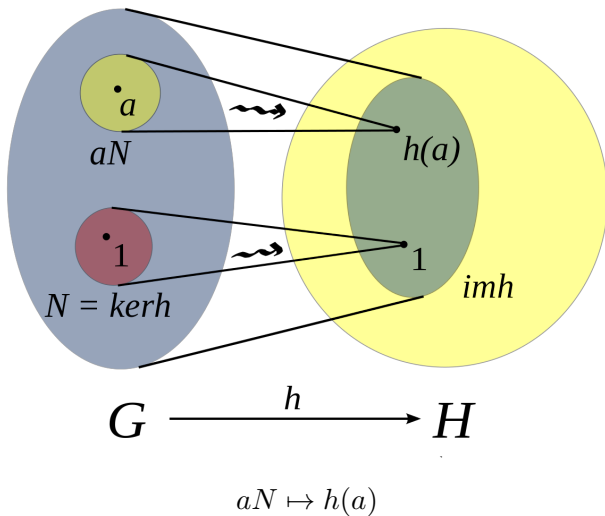
$$G/\text{Ker } \phi \cong \phi(G).$$

同态核可以看作群 G 与其同态像 $\phi(G)$ 之间相似程度的一种度量



$$N = \text{Ker } \phi$$

$$G/(N \triangleq \text{Ker } h) \cong (h(G) \triangleq \text{im } h)$$



$$\phi : \mathbb{Z} \rightarrow \mathbb{R}^*$$

$$n \mapsto (-1)^n$$

$$\text{Ker } \phi = 2\mathbb{Z}$$

$$\mathbb{Z}/(2\mathbb{Z}) = (2\mathbb{Z}, 2\mathbb{Z} + 1) \cong \phi(\mathbb{Z}) = (-1, 1)$$

$$\phi : \mathbb{Z} \rightarrow \mathbb{Z}_6$$

$$a \mapsto [a]_6$$

$$\text{Ker } \phi = 6\mathbb{Z}$$

$$\mathbb{Z}/(6\mathbb{Z}) = \{0 + H, 1 + H, \dots, 5 + H\} \cong \phi(\mathbb{Z}) = \mathbb{Z}_6$$

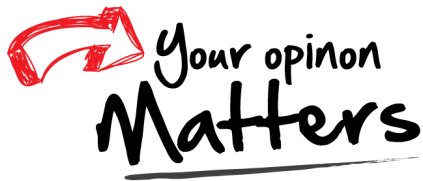
$$\begin{aligned}\phi : \mathbb{R}[x] &\rightarrow \mathbb{R}[x] \\ f(x) &\mapsto f'(x)\end{aligned}$$

$\mathbb{R}[x]$: 全体实系数多项式关于多项式的加法构成的群

$$\text{Ker } \phi = \mathbb{R}$$

$$\mathbb{R}[x]/\mathbb{R} \cong \phi(\mathbb{R}[x]) = \mathbb{R}[x]$$

Thank
You!



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